Alex Pedrini

https://alexpedrini.github.io/
Research Assistant in Astronomy at Stockholm University
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EXPERIENCE

Stockholm University, Stockholm — *Research Assistant*

NOVEMBER 2026 - DECEMBER 2026

Stockholm University, Stockholm — *PhD student*

AUGUST 2022 - NOVEMBER 2026

PhD candidate in astronomy focused on star formation and stellar feedback from emerging young star clusters. Active member of the FEAST program - a JWST survey investigating stellar feedback processes in nearby galaxies.

EDUCATION

Stockholm University, Stockholm — *PhD degree*

AUGUST 2022 - NOVEMBER 2026

PhD degree in Astronomy, defended on November 5th, 2026.

Title: *Young star clusters in the act of emerging: a JWST and HST view of stellar feedback and timescales in the Local Universe*

Opponent: Dr. Edvige Corbelli (INAF-Arcetri)

Supervisors: Angela Adamo (Stockholm University), Arjan Bik (Stockholm University), Daniela Calzetti (UMass Amherst).

Stockholm University, Stockholm — *Licenciate*

AUGUST 2022 - SEPTEMBER 2024

Licentiate's degree in Astronomy, September 26th, 2024; mark: *Pass*

Title: *The emergence phase of star formation as a key probe for feedback timescales*

Supervisors: Angela Adamo (Stockholm University), Arjan Bik (Stockholm University), Daniela Calzetti (UMass Amherst).

Università degli studi di Milano Bicocca, Milano, Italy — *Master's degree*

OCTOBER 2019 - OCTOBER 2021

Master's degree in Astrophysics and Space Physics, October 26th, 2021;

SKILLS

Programming & Scripting: Python, C/C++, Bash

Software & Tools: Astrodendro, SAOImageDS9, CIGALE, SLUG, Jupyter Notebook, TOPCAT, APT and ETC (JWST), PaKo

Web development: Jekyll, GitHub Pages, HTML, Markdown

Operating systems: macOS, Linux, Windows

AWARDS

Stiftelsen Hierta Retzius fond för vetenskaplig forskning, Kungl. Vetenskapsakademien, travel grant, 65000 SEK

2025 Spring Symposium, STScI, travel grant, 1500 USD

Dahlmark 2025, Department of Astronomy, Stockholm University, travel grant, 10000 SEK

Dahlmark 2024 II, Department of Astronomy, Stockholm University, travel grant, 7000 SEK

Dahlmark 2024 I, Department of Astronomy, Stockholm University, travel grant, 5000 SEK

mark: 110/110.

Title: *Towards a complete view of ram pressure stripping in the galaxy cluster A1367*

Supervisors: Michele Fumagalli (UniMiB), Matteo Fossati (UniMiB)

Università degli studi di Milano Bicocca, Milano, Italy — *Bachelor's degree*

OCTOBER 2016 - SEPTEMBER 2019

Bachelor's degree in Physics, September 19th, 2019; mark: 101/110.

Title: *Misure della conducibilità termica di un campione di assorbitore per microonde a temperature criogeniche (Thermal conductivity measurements of a sample of microwave absorber at cryogenic temperatures)*

Supervisors: Mario Zannoni (UniMiB)

TALKS & POSTERS

Talks

Young star clusters in the act of emerging - Invited - JWST online weekly briefing - Aug 20, 2026

Young star clusters in the act of emerging - Invited - PHANGS online colloquium - Jul 14, 2026

The emerging timescale of young star clusters in local galaxies - GalScale, Ascona, Switzerland - Jul 13, 2026

The emerging timescale of young star clusters in local galaxies - Nordic-Baltic Astronomy days, Turku, Finland - May 27, 2026

Young star clusters in the act of emerging - SU PhD seminar, Stockholm, Sweden - May 13, 2026

The emerging timescale of young star clusters in local galaxies - Science with the Hubble and James Webb Space Telescopes VIII, Vienna, Austria - Apr 15, 2026

The emerging sequence of young star clusters in local galaxies - UMASS invited seminar, Amhrest, USA - Nov 12, 2025

The morphological evolution of dusty star-forming regions in NGC 628 - EAS Annual Meeting, Cork, Ireland - Jun 27, 2025

A census of emerging young star clusters in nearby galaxies - EAS Annual Meeting, Cork, Ireland - Jun 24, 2025

Dissecting the SED of emerging young star clusters - STScI Spring Symposium: Inter+Stellar, Baltimore, MD, USA - May 15, 2025

The emergence phase of star formation as a key probe for feedback timescales - SU-UU meeting, Uppsala, Sweden - Nov 1, 2024

(Outreach talk) Stars in the making: a journey through galaxies with the JWST- Astronomins dag och natt 2024, Stockholm, Sweden - Oct 19, 2024

Dahlmark 2023, Department of Astronomy, Stockholm University, travel grant, 10500 SEK

Best Poster Award, Astronomdagarna 2024, Lund Feedback in Emerging extragAlactic Star clusTers: first results

LANGUAGES

Italian (mother language), English (fluent), Swedish (good)

PAH destruction in NGC628 during the emergence phase of star formation - The physics and impact of astrophysical dust, Aspen, CO, USA - Mar 4, 2024

Posters

Feedback in Emerging extragAlactic Star clusTers: first results - Astronomdagarna 2024, Lund, Sweden - Oct 3, 2024

JWST spots PAH destruction from emerging star clusters in NGC 628 - EAS Annual Meeting, Padova, Italy - Jul 1, 2024

The morphology of dusty star-forming regions as revealed by JWST in NGC 628 - Olympian Symposium, Paralia Katerini, Greece - May 29, 2023

MENTORING EXPERIENCE

Alice Knutas - Master Student - 2025

Ella Segerström - Bachelor Student - 2026

SUCCESSFUL OBSERVING PROPOSALS

JWST GO5 - Resolving Stellar Feedback in Action via NIRCam and MIRI Imaging of Very Nearby Galaxies, 2026, 199 hrs (PI: Leroy, A)

JWST GO5 - Feedback in Low-metallicity Environments from Emerging sTar clusters (FLEET), 2026, 64 hrs (PI: Adamo, A)

JWST GO3 - A sharp view on the small-scale star formation in a remarkable redshift 2.58 spiral galaxy, 2024, 16 hrs (PI: Adamo, A)

JWST GO2 - Mapping the rapidly evolving interstellar medium of emerging young star clusters, 2023, 8 hrs (PI: Adamo, A)

JWST GO2 - The JWST Whirlpool Galaxy Treasury, 2023, 62 hrs (PI: Sandstrom, K)

PUBLICATIONS

Corresponding author

1. **Pedrini A.,** Adamo A., Krumholz M. et al., 2026, submitted to ApJ

   The near infrared SED of young star clusters in the FEAST galaxies: an overview from SLUG

2. **Pedrini A.**, Adamo A., Calzetti D., et al., 2026, NatAs, 10, 1179

   The emerging timescale of young star clusters regulated by cluster stellar mass

3. **Pedrini A.**, Adamo A., Bik A., et al., 2025, ApJ, 992, 96P

   The near infrared SED of young star clusters in the FEAST galaxies: Missing ingredients at 1 – 5 μm

4. **Pedrini A.**, Adamo A., Calzetti D., et al., 2024, ApJ, 971, 32P

   Feedback in Emerging extragAlactic Star clusTers: JWST spots polycyclic aromatic hydrocarbon destruction in NGC 628 during the emerging phase of star formation

5. **Pedrini A.**, Fossati M., Gavazzi G., et al., 2022, MNRAS, 511, 5180P

   MUSE sneaks a peek at extreme ram-pressure stripping events - V. Towards a complete view of the galaxy cluster A1367

Others:

1. Bortolini G., ..., **Pedrini A.,** et al., 2026, ApJ, 1012, 41
2. Padave M., ..., **Pedrini A.,** et al., 2026, ApJ, 1009, 175
3. Calzetti D., ..., **Pedrini A.,** et al., 2026, ApJ, 1008, 198
4. Linden S., Adamo A., **Pedrini A.,** et al., 2026, ApJ, 1005, 227
5. Faustino Vieira H., ..., **Pedrini A.,** et al., 2026, ApJ, 1001, 128
6. Lapeer M., ..., **Pedrini A.,** et al., 2026, ApJ, 999, 189
7. Gregg B., ..., **Pedrini A.,** et al., 2026, ApJ, 997, 20
8. Buckner A., ..., **Pedrini A.,** et al., 2026, MNRAS, 545, 3
9. Knutas A., Adamo A., **Pedrini A.,** et al., 2025, ApJ, 993, 13
10. Bortolini G., ..., **Pedrini A.,** et al., 2025, ApJ, 991, 212
11. Calzetti D., ..., **Pedrini A.,** et al., 2025, ApJ, 991, 198
12. Correnti M., ..., **Pedrini A.,** et al., 2025, ApJ, 990, 72
13. Draine B., ..., **Pedrini A.,** et al., 2025, ApJL, 984, L42
14. Elmegreen B., ..., **Pedrini A.,** et al., 2025, OJAp, 8, 21
15. Calzetti D., ..., **Pedrini A.,** et al., 2024, ApJ, 971, 118
16. Gregg B., ..., **Pedrini A.,** et al., 2024, ApJ, 971, 115